

Scanning station: the work cycle — validation and aggregation

This is an informational translation. The matching Russian revision is authoritative. Step-by-step operator guide: scanning codes and station signals, filling and closing boxes, building pallets, working offline, pausing and closing the shift.

Document code	MKR-INS-02
Revision	2026.09/03
Effective from	26 September 2026
Revision verification	https://markiro.app/d/MKR-INS-02/2026.09/03/26.09.2026
Language	EN
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1. Purpose

This instruction covers the operator's work at a Markiro scanning station during a shift: scanning marking codes, station signals, filling and closing boxes in aggregation mode, building pallets, working offline and finishing the shift. Signing in and starting a shift are covered by instruction MKR-INS-01.

The shift mode is set when the shift is created: “Validation” — the station only checks and records every code; “Aggregation” — accepted units are additionally packed into boxes, and the station tracks them and prints box labels; on a shift with pallets it also builds the closed boxes into pallets.

Note: The screenshots use demo data. Product names, shift numbers and counters on your station will differ.

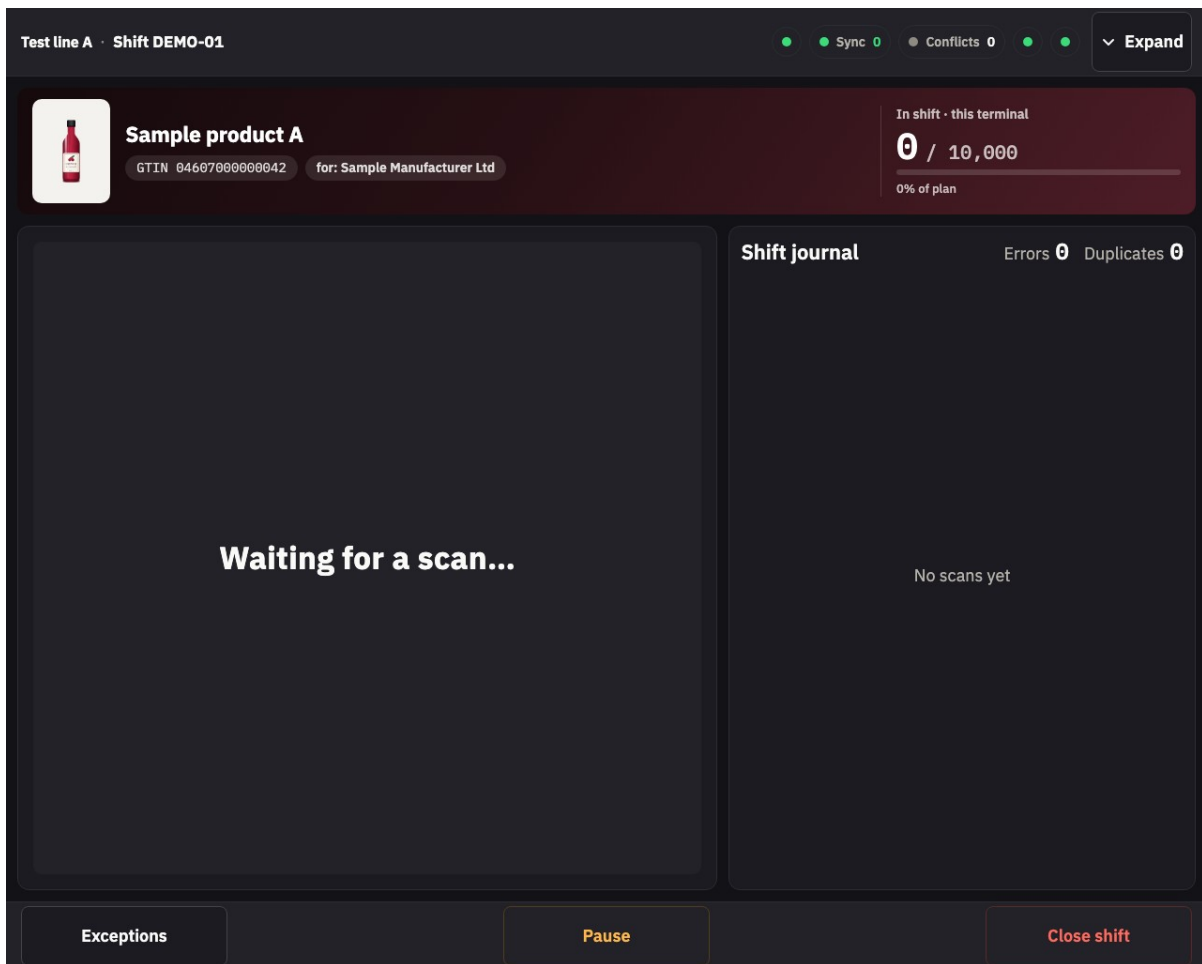
2. The scan cycle and signals

The work screen is laid out as follows: at the top, the shift band with the product (photo, name, GTIN) and the shift total; on the left, the result of the last scan, and in aggregation mode the open box; on the right, the shift journal with this station's latest scans. At the bottom are the “Exceptions”, “Pause” and “Close shift” buttons.

Step 1. Scan a marking code

Take a unit of product and point the scanner at the DataMatrix code. The station processes scans one at a time: wait for the signal for the current unit before scanning the next one.





Work screen: the station is waiting for a scan

Expected result: The station beeped and showed the scan result.

Step 2. Code accepted — keep going

The station shows an accepted scan on the work screen: in validation mode, a green panel with a check mark and the code; in aggregation mode, a check mark with the serial number on the box panel. The “In shift” total on the shift band goes up. In aggregation mode, put the unit into the open box; in validation mode, pass it on down the line.



Test line A · Shift DEMO-01

● Sync 0 ● Conflicts 0 ● Expand



Sample product A
GTIN 04607000000042 for: Sample Manufacturer Ltd

In shift · this terminal
1,248

✓ (01) 04607000000042 (21) DEMO-SERIAL-000128

Shift journal

Errors 2 Duplicates 1

ACCEPTED DEMO-SERIAL-000128	2:32:08 PM
ACCEPTED DEMO-SERIAL-000127	2:32:07 PM
ACCEPTED DEMO-SERIAL-000126	2:32:06 PM
ACCEPTED DEMO-SERIAL-000125	2:32:05 PM
ACCEPTED DEMO-SERIAL-000124	2:32:04 PM
ACCEPTED DEMO-SERIAL-000123	2:32:03 PM

Exceptions

Pause

Close shift

Work screen: the code is accepted (green panel)

Step 3. The “DUPLICATE” signal — set the unit aside

This code has already been scanned: the station shows the time of the first scan. Do not put the unit into a box — set it aside separately. If duplicates keep coming, stop and tell your supervisor: the product may have already passed through the station.



**DUPLICATE**

First scanned at 2:31:52 PM

The “DUPLICATE” signal with the time of the first scan

Step 4. A red signal — do not let the unit through

“WRONG CODE” — the scan was not a marking code or the code is damaged: flatten the packaging and scan again; if the code cannot be read, set the unit aside. “WRONG GTIN” — the product does not belong to this shift: remove it from the line. “WRITE FAILED” — the station could not save the scan: stop and call a technician.





WRONG GTIN

A red signal: the unit must not pass

Note: The full-screen signal fades out by itself after a few seconds. If you got distracted, check the “Shift journal” panel: it lists the latest scans with the verdict, serial number and time.

3. “Validation” mode

In validation mode the operator's job is to run every unit through the scanner and watch the signals. There are no boxes and no printing in this mode.

- The shift band shows the shift progress: “In shift · all terminals” is how many units all terminals have accepted in the shift. If a plan is set, the plan follows the total, with a bar and the share completed below, for example “14% of plan”. If other terminals work the shift too, this station's share is added — “this terminal 302”.
- The station shows the “In shift · this terminal” label until it has received the total from the server — for example, when the shift was opened without a connection: the number then counts only its own scans. If the server total has not been refreshed for more than two minutes and other terminals work the shift, the time of the last answer replaces this station's share — “other terminals as of 11:58 AM”.
- The “Shift journal” panel shows this station's latest scans: the verdict, serial number and time. The GTIN is shown only on “WRONG GTIN” rows; undoing the last scan adds an “Undone” row.
- The “Errors” and “Duplicates” counters in the journal header count this station's rejected scans for the whole shift and are not reset by a pause, an operator change or a station restart.
- When this station's own scans reach the plan, it reports “Plan completed” — follow your supervisor's directions from there.



4. “Aggregation” mode

Step 1. Fill the open box

The box panel shows the box number (“Box no. 1”), how full it is — how many items are already in the box out of its capacity — and the serial number of the last accepted code with a check mark. Put a unit into the box only once the check mark with its serial number has appeared on the panel.

The screenshot displays the 'Aggregation' mode interface. At the top, it shows 'Test line A · Shift DEMO-01' and status indicators for Sync (0), Conflicts (0), and an Expand button. Below this is a header for 'Sample product A' with a GTIN (84607000000042) and manufacturer (Sample Manufacturer Ltd). A counter shows 'In shift · this terminal' with a value of 1,248.

The main section is divided into two parts. On the left, the 'Box no. 1' panel shows a progress bar for box 1 out of 10. The progress bar has 10 segments, with the first two filled green and the third outlined in blue. Below the progress bar are buttons for 'Close box', 'Undo last scan', and 'Clear box'. On the right, the 'Shift journal' table lists accepted items with their serial numbers and timestamps.

Box no.	Items	Serial Number	Timestamp
1	1	✓ DEMO-SERIAL-000128	2:32:08 PM
1	2	✓ DEMO-SERIAL-000127	2:32:07 PM
1	3	✓ DEMO-SERIAL-000126	2:32:06 PM
1	4	✓ DEMO-SERIAL-000125	2:32:05 PM
1	5	✓ DEMO-SERIAL-000124	2:32:04 PM
1	6	✓ DEMO-SERIAL-000123	2:32:03 PM

At the bottom, there are buttons for 'Exceptions', 'Pause', and 'Close shift'.

Aggregation work screen: the open box panel

Step 2. The station closes a full box by itself

When the last item that fits the box's capacity goes in, the station closes the box by itself: it assigns the box an SSCC number and sends the label to the printer. Stick the label onto this box right away — do not put it aside. The “Close box” button closes a box earlier — for example, a partially filled box at the end of the shift. If the panel says “Capacity not set”, the station cannot tell when a box is full, so close every box with this button.



Test line A · Shift DEMO-01

● Sync 0 ● Conflicts 0 ● Expand



Sample product A
GTIN 84607000000042 for: Sample Manufacturer Ltd

In shift · this terminal
1,248

Box no. 1 **120** / **120** Items ✓ DEMO-SERIAL-000128

One cell represents several positions

Close box

Undo last scan

Clear box

Shift journal

Errors 2 Duplicates 1

ACCEPTED DEMO-SERIAL-000128	2:32:08 PM
ACCEPTED DEMO-SERIAL-000127	2:32:07 PM
ACCEPTED DEMO-SERIAL-000126	2:32:06 PM
ACCEPTED DEMO-SERIAL-000125	2:32:05 PM
ACCEPTED DEMO-SERIAL-000124	2:32:04 PM
ACCEPTED DEMO-SERIAL-000123	2:32:03 PM

Exceptions

Pause

Close shift

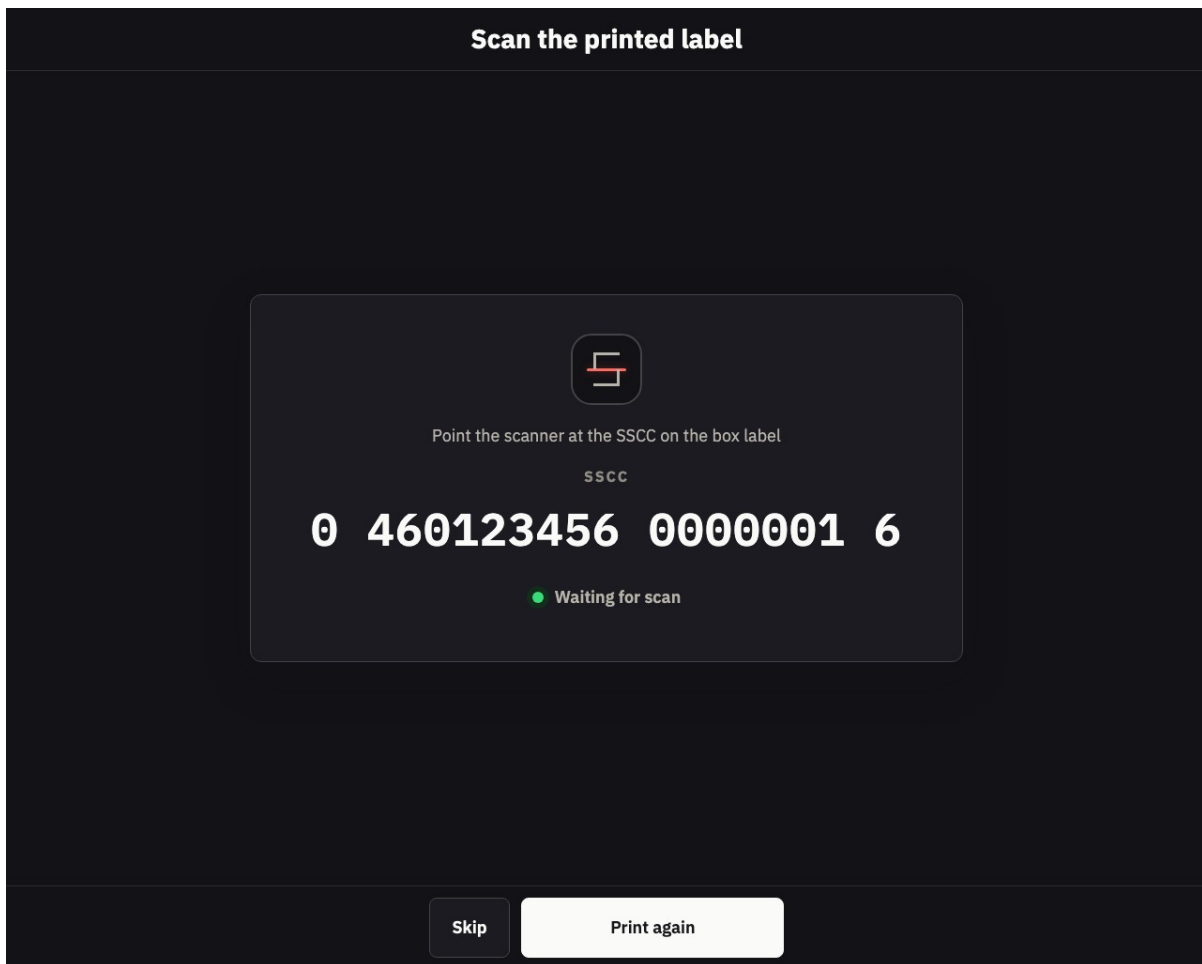
The box is full: the station closes it and prints the label

Expected result: The printer printed the box label.

Step 3. Verify the printed label

If print verification is enabled on the station, the “Scan the printed label” screen appears: point the scanner at the SSCC code on the label. The “This is a different label” message means you are holding a label from another box — find the right one or tap “Print again”. The “Skip” button skips the verification; the skip is recorded.





Verifying the printed box label

Expected result: The station confirmed the label and opened the next box.

Important: Every label belongs to one specific box: the SSCC number is unique. A label stuck onto the wrong box breaks the records of the whole batch.

Note: “Undo last scan” removes the last added item from the open box, and “Clear box” removes all of its items — use them only when your supervisor says so.

5. Pallets

If pallet assembly is on for the shift, the station puts every closed box onto the current pallet right away, and when the pallet is full it closes it and prints its label. Pallet assembly is switched on when the shift is planned in the cabinet, or with “With pallets” when an “Aggregation” shift is created at the station. The number of boxes per pallet comes from the product card; it can be changed when the shift is planned in the cabinet.

Step 1. Watch the pallet strip

Under the box panel is the “Pallet” strip: how many boxes are already on the pallet out of its capacity (for example, 15 / 66) and how full it is in percent. Below are the boxes still to go (for example, “51 boxes remaining”) and, after the word “last”, the end of the SSCC of the last box put on it. The “Pallet contents” button lists the current pallet's boxes, and “Close pallet” closes it early; both are unavailable while the pallet has no boxes.



Test line A · Shift DEMO-01

● Sync 0 ● Conflicts 0 ● Expand



Sample product A
GTIN 84607000000042 for: Sample Manufacturer Ltd

In shift · all terminals

1,302 / 9,580

14% of plan · this terminal 302

Box no. **1** / **20** Items

✓ DEMO-SERIAL-000128

Close box

Undo last scan

Clear box

Pallet **15** / **66** boxes · 23%
51 boxes remaining · last ...619998

Pallet contents

Close pallet

Exceptions

Pause

Close shift

Shift journal

Errors **2** Duplicates **1**

ACCEPTED DEMO-SERIAL-000128

2:32:08 PM

ACCEPTED DEMO-SERIAL-000127

2:32:07 PM

ACCEPTED DEMO-SERIAL-000126

2:32:06 PM

ACCEPTED DEMO-SERIAL-000125

2:32:05 PM

ACCEPTED DEMO-SERIAL-000124

2:32:04 PM

ACCEPTED DEMO-SERIAL-000123

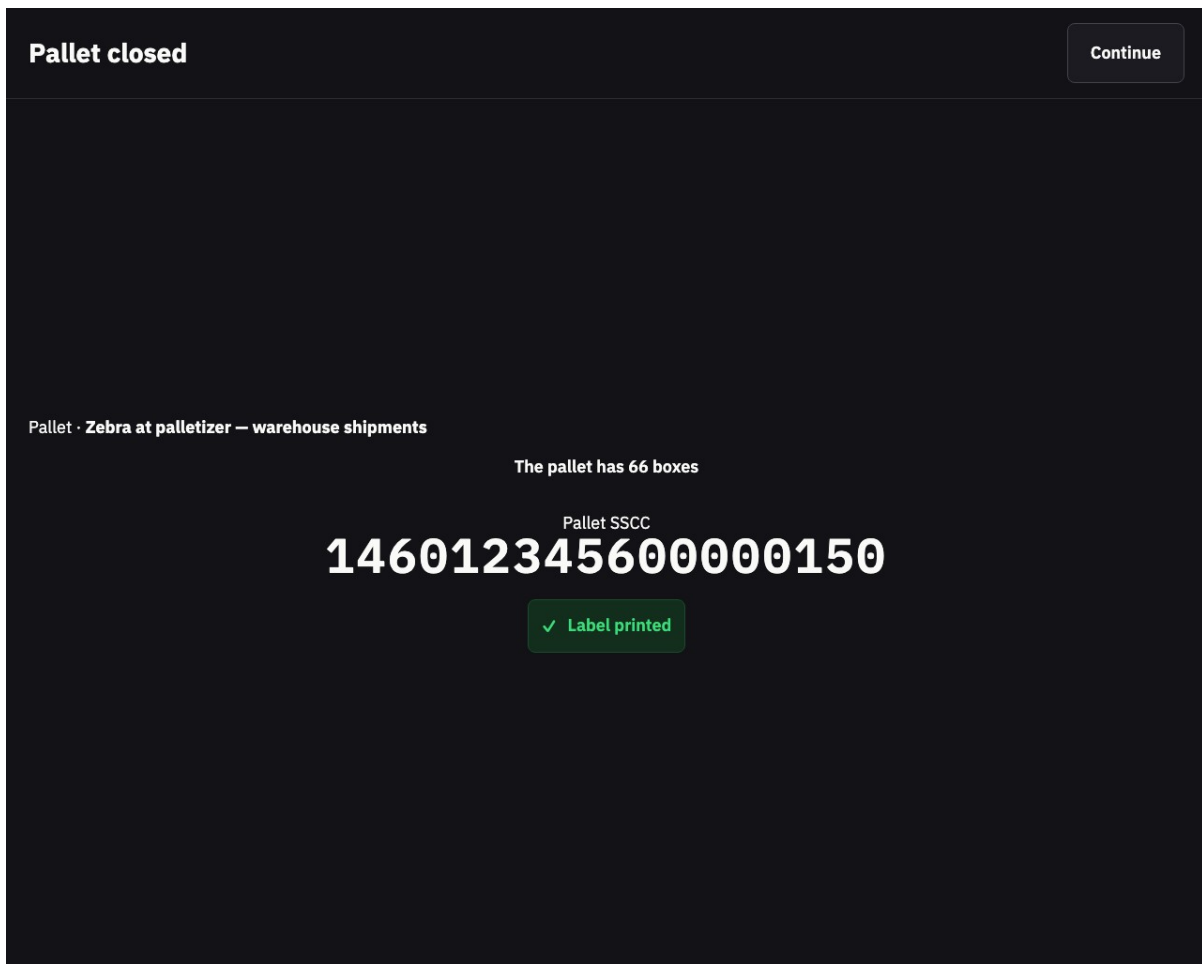
2:32:03 PM

The work screen of a pallet shift: the pallet strip under the box

Step 2. The station closes a full pallet by itself

When the box that fills the pallet is closed, the station closes the pallet too: it assigns it an SSCC number and prints the label. The “Pallet closed” screen shows the box count, the pallet SSCC and the printer the label goes to; while printing, it says “Printing the pallet label...”. Stick the label onto this pallet and tap “Continue” — the next boxes go onto a new pallet.





The pallet is closed and its label printed

Expected result: The “Pallet closed” screen shows “Label printed”.

Step 3. If the pallet label did not print

If printing fails, the station shows the cause and a “Retry printing” button, plus “Set up printer” when the cause is the printer. If printing was interrupted — for example, the station restarted mid-print — the station cannot tell whether the label came out and shows “Unknown whether the label printed”. It never prints again by itself: check the printer and tap “The label already printed” if the label came out, or “Print again” if it did not. The “Change printer” button sends the label to another printer of the station.



Pallet closed

The label already printed

Pallet · Zebra at palletizer — warehouse shipments

Change printer

The pallet has 66 boxes

Pallet SSCC

146012345600000150

Unknown whether the label printed

Print again

The print outcome is unknown: confirm it or print again

Important: If printing is impossible right now, “Continue without a label” returns to work: the pallet is already closed and recorded — print its label later with “Reprint pallet” (the exceptions instruction MKR-INS-03). Mark such a pallet so it does not get lost.

Step 4. Check the pallet contents

“Pallet contents” lists the boxes of the current pallet: each box's SSCC and closing time. While the list is open, scanning is paused; the “Back to assembly” button returns to work.



Pallet contents	
The pallet has 15 boxes	
Box SSCC	Closed at
004601234560619998	9/12/26, 2:42:00 PM
004601234560619997	9/12/26, 2:41:00 PM
004601234560619996	9/12/26, 2:40:00 PM
004601234560619995	9/12/26, 2:39:00 PM
004601234560619994	9/12/26, 2:38:00 PM
004601234560619993	9/12/26, 2:37:00 PM
004601234560619992	9/12/26, 2:36:00 PM
004601234560619991	9/12/26, 2:35:00 PM
004601234560619990	9/12/26, 2:34:00 PM
004601234560619989	9/12/26, 2:33:00 PM
004601234560619988	9/12/26, 2:32:00 PM
004601234560619987	9/12/26, 2:31:00 PM
004601234560619986	9/12/26, 2:30:00 PM
004601234560619985	9/12/26, 2:29:00 PM
Back to assembly	

Pallet contents: the boxes and their closing times

Step 5. Close a pallet early

To close a pallet that is not full — for example, at the end of a batch — tap “Close pallet”. The station asks to “Close pallet early” and shows how many boxes the pallet has; tap “Confirm” or “Stay”. The rest is the same as for a full pallet: an SSCC number and the label print.

Important: “no pallet serials left” on the pallet strip means the station has run out of SSCC numbers for pallets: a full pallet does not close, and the next boxes go onto it beyond capacity. Restore the station's connection to the server (call an administrator) — the station receives new numbers and closes the pallet when the next box is closed; after that you can also close it with “Close pallet”.

6. Working offline

Step 1. Keep working during “No connection”

If the “Server” indicator in the status bar shows “No connection”, the station keeps accepting scans and stores them locally — the “Sync” indicator shows how many scans are waiting to be sent. Work as usual: once the connection is back, the data goes to the server by itself. Meanwhile the “In shift” total counts the other terminals' scans as of the server's last answer (see section 3).



Test line A · Shift DEMO-01

Server No connection Sync 7 Conflicts 0 Expand

 **Sample product A**
GTIN 04607000000042 for: Sample Manufacturer Ltd

In shift · this terminal
1,248

✓ (01) 04607000000042 (21) DEMO-SERIAL-000128

Shift journal Errors 2 Duplicates 1

ACCEPTED DEMO-SERIAL-000128	2:32:08 PM
ACCEPTED DEMO-SERIAL-000127	2:32:07 PM
ACCEPTED DEMO-SERIAL-000126	2:32:06 PM
ACCEPTED DEMO-SERIAL-000125	2:32:05 PM
ACCEPTED DEMO-SERIAL-000124	2:32:04 PM
ACCEPTED DEMO-SERIAL-000123	2:32:03 PM

Exceptions Pause Close shift

Working offline: scans accumulate on the station

Expected result: After the connection is restored the number on the “Sync” indicator goes down to zero.

Important: Call an administrator if the number on the “Sync” indicator does not go down for a long time while the network is up, or the indicator shows “Not syncing”. In aggregation mode a long offline period can exhaust the box number reserve — the station will report “Box numbers have run out” and pause scanning until the connection is back.

7. Pausing and closing the shift

Step 1. Take a break with “Pause”

The “Pause” button takes you out of the shift without closing it: the station returns to the shift selection screen, and after the break you or another operator can continue the shift. If some scans have not reached the server yet, the station warns about it and offers “Stay” or “Leave anyway”; the data stays on the station and is sent once there is a connection.

Step 2. Close the shift at the end of work

Tap “Close shift”. In aggregation mode close the open box first — the station reminds you: “Close the open box first”. On a pallet shift, if the current pallet has boxes, the station asks “The pallet has 15 of 66 boxes — close it?”: “Close pallet and shift” closes the pallet and prints its label as in section 5 — after “Continue” on the “Pallet closed” screen the station closes the shift; “Stay” returns to work. If pallet numbers have run out, the station reports “Pallet numbers have run out” — restore the connection and try again. If the actual quantity does not match the plan, the station asks for the reason for the difference.

Expected result: The shift is closed and the station returned to the shift selection screen.



8. Common problems

Duplicates keep coming one after another — This product has probably been scanned before. Stop, set the batch aside and call your supervisor.

The printer did not print the box label — The station shows the “The box label was not printed” screen with options to reprint or continue without a label. Print failure handling is covered in detail by the exceptions instruction.

The pallet label is damaged or lost — Reprint it: “Exceptions” → “Pallet actions” → “Reprint pallet” (the exceptions instruction MKR-INS-03). Destroy the old label.

The pallet does not close although it is full — The pallet strip shows “no pallet serials left”: the station has run out of pallet numbers. Restore the station's connection to the server (call an administrator); see section 5 for details.

Box numbers have run out — The SSCC number reserve was exhausted while offline. Restore the station's connection (call an administrator), then get back to work.

The number on the “Sync” indicator grows although the network is up — Do not stop working — scans are not lost. Tell an administrator: the send queue needs attention.

The scanner stopped reading codes — Check the scanner indicator in the status bar — a problem indicator expands into a caption, and the “Expand” button opens the full view — and check the cable. If the connection does not come back, call a technician.

If your problem is not listed above, contact your organization's administrator or Markiro support: hello@v-b.tech.

